

Standard-Slope Integration (SSI) vs. RKF45 — Long-Duration Accuracy Comparison

Standard-Slope Integration (SSI) is a slope-defined numerical integration operator that reconstructs system behavior without step accumulation. Unlike traditional integrators, SSI operates directly on derivative information, producing globally coherent trajectories that preserve geometric invariants with machine-precision stability.

This report compares SSI against the classical Runge–Kutta–Fehlberg (RKF45) method over 10, 20, 30, and 40 full revolutions of circular motion. Four metrics were evaluated:

- Radius Error
- Position Error
- Area Error
- Angle Drift (deg)

SSI Results

SSI maintains exact geometric coherence across all revolutions:

- Radius error: 0
- Position error: 0
- Angle drift: 0
- Area error: $\sim 10^{-21}$ (approximately one hundred million times smaller than RKF45)

SSI's slope-based reconstruction avoids the geometric drift inherent to step-based numerical methods. Even after 40 full revolutions, SSI returns to the exact starting point with no radius loss, no angular bias, and area preservation at the sextillionth scale.

RKF45 Results

RKF45 performs as expected for a step-based integrator:

- Radius error: approximately -10^{-13} (slightly negative, indicating radius loss)
- Position error: grows in the 10^{-13} range
- Area error: increases from approximately -10^{-13} to nearly -4×10^{-13}
- Angle drift: accumulates from approximately 10^{-11} to about 4×10^{-11} by 40 revolutions

Radius, Position, and Area Error Summary					
Method	Rev	Radius Error	Position Error	Area Error	Angle Drift (deg)
SSI	10	0	0	7.77E-21	0
	20	0	0	7.73E-21	0
	30	0	0	7.89E-21	0
	40	0	0	7.89E-21	0
RKF	10	-1.42×10^{-13}	2.70×10^{-13}	-1.15×10^{-13}	1.32×10^{-11}
	20	-1.71×10^{-13}	4.60×10^{-13}	-2.14×10^{-13}	2.45×10^{-11}
	30	-2.21×10^{-13}	6.27×10^{-13}	-2.93×10^{-13}	3.36×10^{-11}
	40	-1.47×10^{-13}	6.97×10^{-13}	-3.41×10^{-13}	3.90×10^{-11}

SSI maintains exact radius and position closure and preserves area to the 10^{-21} level, while RKF45 exhibits the expected cumulative drift of step-based numerical integration.

RKF45 was run in its classical fixed-step form without adaptive step sizing or tolerance control. Adaptive RKF45 was intentionally not used because adaptive metrics reduce local truncation error but cannot eliminate the global geometric drift—radius contraction, area loss, position drift, and angle bias—that step-based integrators inherently accumulate.

These values reflect normal floating-point accumulation inherent to incremental propagation.

Conclusion

SSI operates in a structurally stable numerical regime, preserving geometric invariants with machine-precision accuracy over arbitrarily long trajectories. Unlike step-based integrators, SSI does not accumulate radius loss, area drift, position error, or angular bias. RKF45 performs well for a classical method, but its cumulative geometric drift highlights the inherent limitations of incremental propagation.

Beyond long-duration trajectory monitoring, SSI's simplicity, ease of implementation, and invariant-level accuracy make it broadly useful in practical computation. SSI provides a fundamentally different alternative for any application requiring stable integration, high-fidelity reconstruction from derivative information, or long-duration numerical reliability. Its slope-defined formulation enables accurate, drift-free behavior across geometric, physical, dynamical, and analytical systems, offering both researchers and practitioners a robust replacement for traditional step-based schemes.